

Clean Air & Climate Resilience

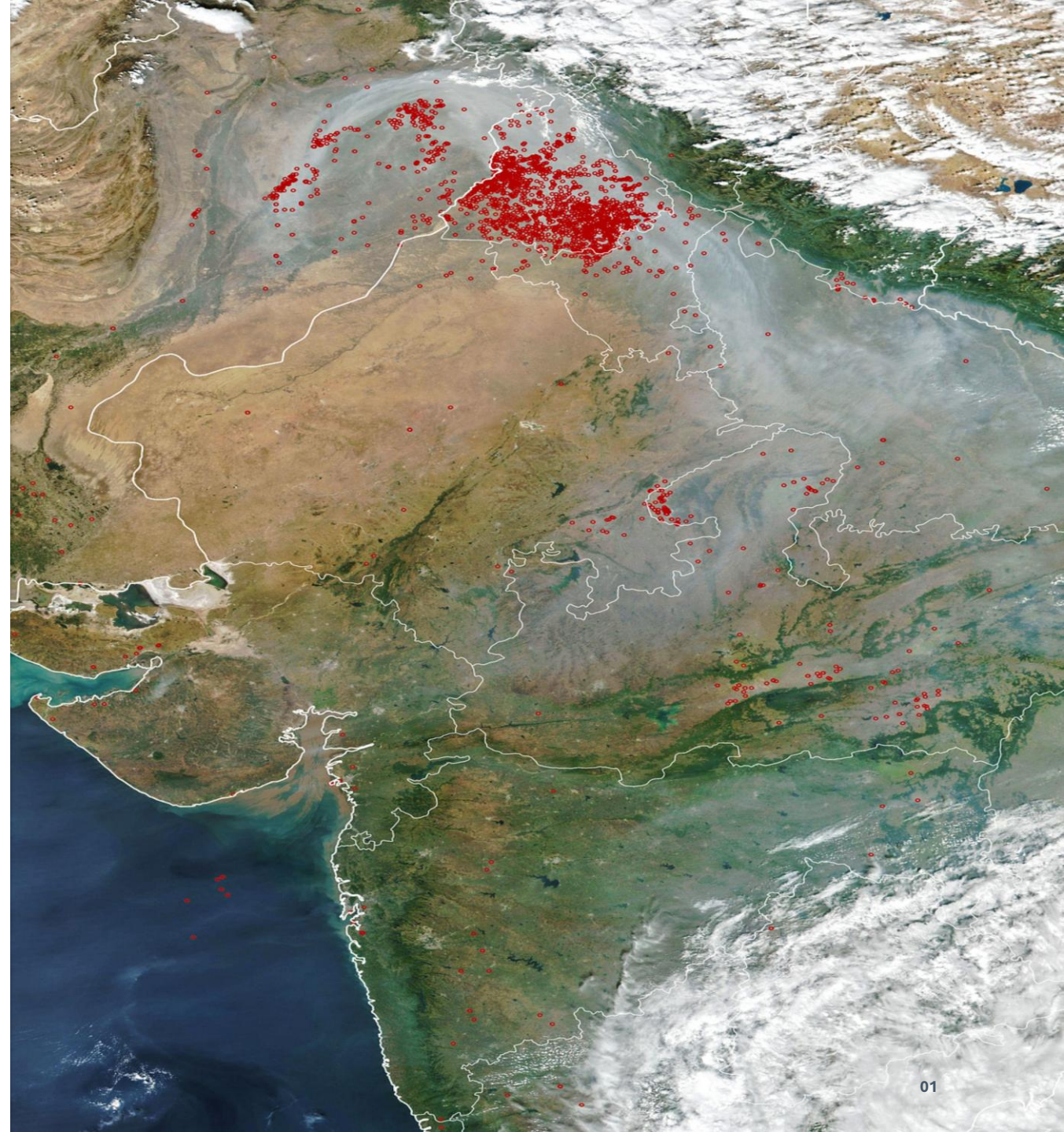
AI-powered detection of hidden pollution hotspots during Punjab–Haryana's stubble-burning season.

DETECT → FORECAST → ACT

SATELLITE DATA

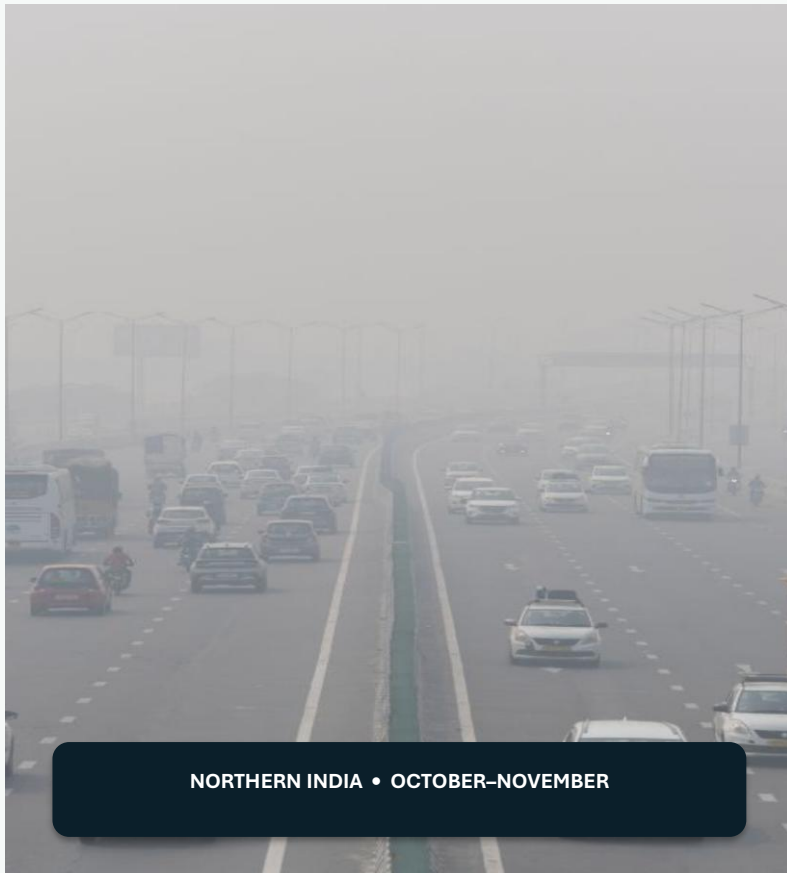
CITIZEN PHOTOS

GROUND STATIONS



Pollution hotspots stay invisible until it's too late

A seasonal crisis is amplified by sparse monitoring and reactive intervention.



01 • SEASONAL CRISIS

Stubble burning drives severe smog

Punjab–Haryana burning can blanket Northern India in toxic haze and push AQI into severe conditions.

02 • SPARSE MONITORING

Hyper-local hotspots go undetected

Ground-station networks are thin and unevenly distributed, leaving communities without timely warning.

03 • REACTIVE, NOT PREDICTIVE

Intervention arrives after the spike

Without forecasting capability, authorities respond after exposure has already increased.

Fuse three real signals into one early-warning system

Independent data streams fill each other's blind spots and feed a weighted hotspot score.

01

SATELLITE DATA

Live Sentinel-5P aerosol index from Copernicus Data Space provides wide-area atmospheric coverage.

WIDE-AREA COVERAGE

02

CITIZEN PHOTOS

Gemini Vision classifies smog and haze severity from resident-submitted images.

GROUND-TRUTH LAYER

03

GROUND STATIONS

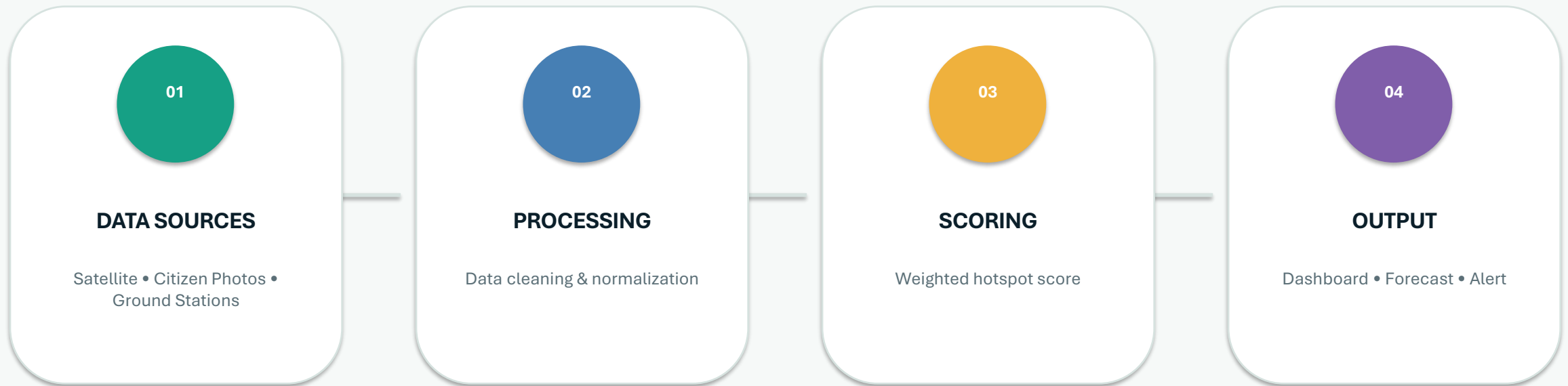
9 real OpenAQ stations contribute 7,489+ pollutant readings to anchor local measurements.

LOCALIZED ANCHOR

WEIGHTED HOTSPOT SCORE → 3-DAY PM2.5 FORECAST → PLAIN-LANGUAGE AI ALERT

How data flows through the pipeline

A modular, near-real-time chain from raw ingestion to actionable output.



FEDERATED BY DESIGN

Punjab–Haryana and Delhi–NCR run independent pipelines; no shared raw data is required between regions.

Technology Stack

A lightweight Python stack combining AI, geospatial processing, real air-quality data, and a deployable web backend.

APPLICATION

- Python**
Core runtime
- Flask 3.1.0**
Web backend
- Flask-CORS 6.0.5**
Cross-origin API
- python-dotenv 1.0.1**
Configuration

AI + DATA

- Google GenAI 2.18.0**
Gemini integration
- Pandas 2.2.3**
Data processing
- NumPy 2.1.3**
Numerical computing
- Scikit-learn 1.5.2**
ML / scoring
- SciPy 1.17.1**
Scientific computing
- Joblib 1.5.3**
Model utilities

GEO + VISION

- SentinelHub 3.11.5**
Sentinel-5P
- Shapely 2.1.2**
Geospatial geometry
- PyProj 3.7.2**
Coordinate systems
- UTM 0.9.0**
Geo conversion
- Pillow 12.3.0**
Image handling
- Requests 2.32.3**
HTTP data access

DATA & DEPLOYMENT • Copernicus Data Space • OpenAQ • Gemini API • SQLite • Render

Selected from the project's actual Python environment — production dependencies only.

Built entirely on free-tier infrastructure

Every component was selected to run without a credit card.

AVOIDED • REQUIRES BILLING

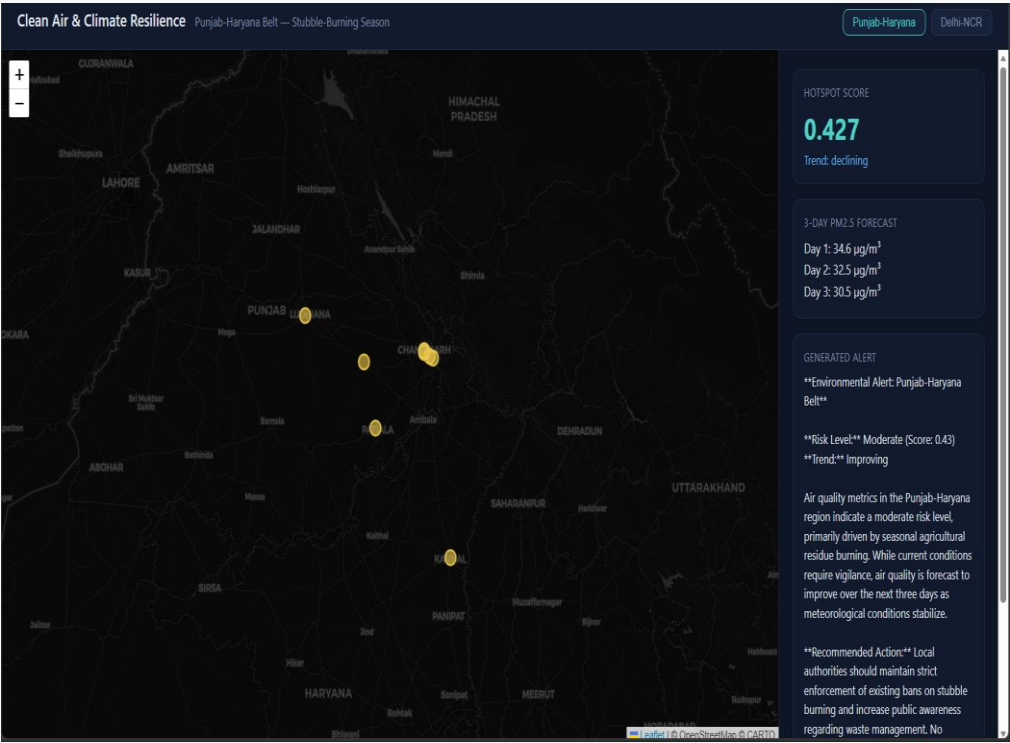
- × Google Earth Engine
- × Vertex AI
- × Firestore / BigQuery
- × Google Cloud Run

USED INSTEAD • FREE / NO CARD

- ✓ Copernicus Data Space (sentinelhub) — live Sentinel-5P data
- ✓ Gemini API via AI Studio — vision + alert generation
- ✓ SQLite — lightweight file-based storage
- ✓ Render — free-tier app hosting

Real station data, live on a two-region dashboard

The demo combines real OpenAQ readings, a computed hotspot score, a 3-day PM2.5 forecast and an AI-generated alert.



LIVE MAP

Station markers across Punjab–Haryana and Delhi–NCR



HOTSPOT SCORE

Weighted composite score from all three data streams



3-DAY FORECAST

PM2.5 trend prediction powered by the fused model



AI ALERT PANEL

Plain-language Gemini alert for local decision-makers

REAL DASHBOARD • LIVE MAP + FORECAST + AI ALERT

What's real vs. simulated

Transparency is part of the architecture: live capabilities are separated from integration-ready components.

REAL • LIVE & COMPUTED

- Satellite data — live pull from Copernicus Data Space
- Citizen photo classification — Gemini Vision tested on sample images
- Weather & AQI history — 9 stations, 7,489+ real OpenAQ readings
- Forecast, hotspot score & alert — computed end-to-end in real time

SIMULATED • ARCHITECTURE READY

- Live photo intake pipeline — supported, not yet wired to a live source
- Cross-border data sharing — simulated with a two-region mock
- Federated architecture is already in place for expansion

The simulated pieces are integration placeholders — the pipeline is ready to scale.

Built to scale across state and national boundaries

The same pipeline can run independently on any region bounding box.

1

ZERO-COST STACK

Any resource-constrained government, NGO, or student team can deploy without enterprise licensing.

2

FEDERATED BY DESIGN

No shared raw data is required between regions, supporting data sovereignty and parallel deployment.

3

EXTENSIBLE ANYWHERE

Any pollution-prone region with satellite coverage and citizen participation can adopt the system.



Data sources & references

Open data and free-tier APIs keep the project reproducible.



SATELLITE DATA

[Copernicus Data Space Ecosystem](#)

dataspace.copernicus.eu



GROUND-STATION AIR QUALITY

[OpenAQ open data platform](#)

openaq.org



AI CLASSIFICATION & ALERTS

[Google Gemini API via AI Studio](#)

ai.google.dev



HOSTING & DEPLOYMENT

[Render free-tier hosting](#)

render.com

LOCKED STUDY REGION

Punjab–Haryana belt • bounding box [75.5, 29.5, 77.5, 31.5] • season: Oct 15 – Nov 15

Thank You

AI, open data, and civic engagement can work together to protect communities before the crisis hits.

FROM DETECTION → FORECAST → ACTION

GITHUB • [sushantgarde/Clean-air-resilience](https://github.com/sushantgarde/Clean-air-resilience)

LIVE DEMO • clean-air-resilience.onrender.com

